

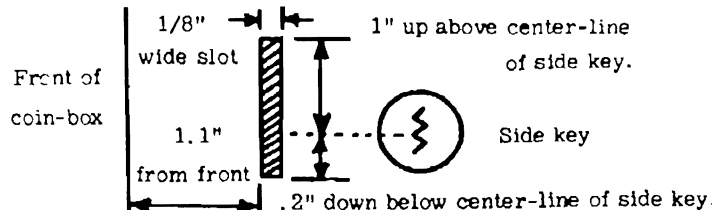
## MORE FORTRESS FUN!

Last week, I met with "CJ" and "The Professor" and told them how much I would like a Fortress Fone of my very own to play with. About 1/2 hour after they left, they returned with one and gave it to me to open. "The Professor" was able to pick the upper lock after removing the shroud around it. It's only 4 levers. I immediately went to work on it while "CJ" bashed open the coin box. After he opened it, we made the following discoveries:

1. The drill pattern you published in #30 for the front plate would make it almost impossible to get the box open, because the bolts go thru double sets of slots before going into the outer shell of the phone. When the rotary plate is drilled as in #30, it should rotate and retract the bolts.

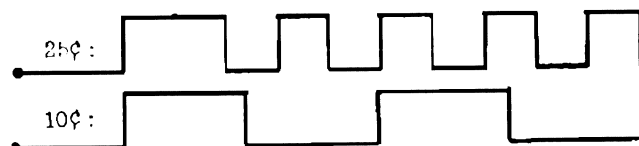
2. The right-hand bolt is really wide and protects the side lock (interlock).

3. The following cutting pattern would be better than a drilling pattern providing you have access to the side:



The cut should be 1.2" long, 1/8" wide, and 1" deep. It can be done with a special grindstone drill bit, and will shear off the interlock mechanism as well as cut the spiral front plate free so it can turn. We are checking reports of a puller for the cover. If there is such an animal, and if it does work, you'll hear about it first in TAP!

4. The tones' intervals are not generated by the coins pushing levers on the red box mechanism (like everyone thinks) but by a little relay which reverses the direction of the wheel that the levers cock. In other words, when a coin is dropped, the coin pushes a lever which cocks a wheel. The coin then drops into the drop chute, and the red box goes thru a 70ms delay to allow the coin to clear the mechanism. After that, a unijunction-timer circuit pulls in a relay which unwinds the wheel until it is back where it started. The number of times the relay ticks, pulling the wheel, is the number of beeps. If the wheel went 5 notches forward, a switch drops, telling the unit to do 35ms beeps (for a quarter). However, the first bleep will always be the same as the above-mentioned 70ms pre-bleep delay, because the coin will just be clear, and all of the pulses (beeps) are actually symmetrical square waves:



I will restate the specs on these tones because they are very important and must be known by all TAP readers. The 25¢ tones are 70ms on, 35ms off, 35ms on, 35ms off, 35ms on, 35ms off, 35ms on, 35ms off, 35ms on. The dime is 80ms on, 80ms off, 80ms on. These can be fudged to 70ms if needed, and better be generated digitally, because they'll fuck up with analog timers, "windows", and analog delays.

Then I made a few discoveries myself:

1. Coin Drop Mechanism - In my last letter, I asked about the "coin-sense relay" but couldn't figure how Bell knew about the money. Now I know! First, they use a low current (20ma) high voltage (50v) relay to release the money. Second, the relay resets a vane which senses the presence of the money. This vane cannot be checked by the operator. However, she can see the current drain caused by the drop relay pulling in. If the vane hasn't been hit by money, the coin-drop relay won't pull in even when the operator tries to make it. The good part is that she can't check the vane until she drops the money. So it doesn't matter when you drop your nickel, as long as you drop one before each time she collects. Also, she can't check it on



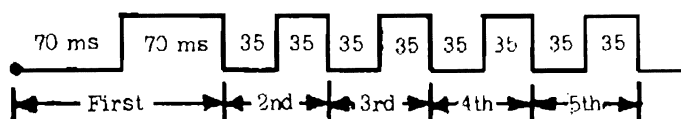
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the initial three minutes because that drops automatically, and so you can talk for at least three minutes for free.

2. The pay phone red box tones drift with the outside temperature. So...before boxing away, make sure the pay-phone frequencies match your red box frequencies or you'll get screwed when you drop your 5¢.

3. As I told you last time, the 25¢ set of tones are 5 beeps, the first one twice as long as the rest:



All of the above info is confirmed by an in-depth experiment which can be done on any phone without taking it out of the booth!

1. Drop 5¢ into a pay phone. Listen to the side and hang up. The relay will pull the wheel back once and click.

2. Drop two nickels into the phone, one at a time. Listen to the receiver or the side of the phone. When the second nickel is dropped, two clicks of the relay are heard - as if a dime had been dropped.

3. Now for some fun: Drop a nickel, and then a dime. There are three clicks! Amazing, a 15¢ piece!

4. Now try dropping two dimes right together and you'll hear both dimes hit the bottom of the coin holding chute before you hear four clicks! a 20¢ piece!

5. Try a nickel and a quarter and get six clicks (30¢).

6. If you're fast enough, this will work too: Drop a nickel. Then two dimes together and you'll hear the familiar quarter sequence and not five nickel beeps.

Whether you are listening to the beeps thru the receiver or clicks on the side of the phone, you'll find these experiments interesting.

If you don't believe me about the timing of the tones on the 25¢ beeps, try tape recording them (have a friend call you from a pay phone and drop money) at 7 1/2 ips on a reel-to-reel recorder. Slow down the tape on playback to as slow as possible and mark the tape at the beginning and end of each bleep. When done marking, measure the marks and you can determine the tone durations. You will find most phones adjusted perfectly, whether single tone or mf phones, so the first bleep is twice as long as the rest.

If you should ever have access to a phone in a relatively private place, and wish to record the tones interference-free, get a phone pickup coil from someplace like Radio Shack (blech!) and place it 3" behind the top keyhole on the right side of the Fortress Fone. It picks up the tones fine there.

Computer Wizard

Some Fortresses must have a mechanism defect. It seems that they will take two dimes or two nickels - a 50¢ savings! Please advise readers to deposit two nickels first and listen for second dial tone. This is for a 20¢ local call area.

Mugs

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# THE PRINCIPLES OF TWX PHREAKING

by  
Cheshire

## INTRODUCTION

IN 1971, THE WESTERN UNION TELEGRAPH CO. BROUGHT THE TELETYPE/TELETYPE EXCHANGE (TWX) NETWORK FROM THE AMERICAN TELEPHONE & TELEGRAPH CO. UNDER THIS AGREEMENT, AT&T COULD NOT ENTER DATA COMMUNICATIONS UNDER 200 BAUD (20 CHARACTERS PER SECOND) UNTILL APRIL 1982. MEANWHILE, WHY WOULD ANY OF THE INDEPENDENT TELCOS BUILD WHOLE NEW EXCHANGES FOR THIS PR. BELL PLAYTOY? THEY WOULDN'T. OF COURSE, THEY TRANSLATE FROM FUNNY TWX AREA CODES TO NORMAL TELEPHONE EXCHANGES. FIND THE RIGHT EXCHANGE, AND WIN THE TWX NETWORK.

## TRANSLATING TWX EXCHANGES

THERE IS AN EASY WAY OF TRANSLATING A TWX EXCHANGE. THAT'S WHEN IT IS "GIVEN" TO YOU IN THE TWX DIRECTORY. TWX AREA CODES ARE USUALLY 510, 710, 810, & 910 IN THE UNITED STATES, AND 610 IN CANADA. SOME TWX MACHINES HAVE REGULAR TELEPHONE NUMBERS, BECAUSE THEY WERE INSTALLATIONS BEFORE THE MODEL 32 TELETYPE BECAME THE STANDARD OF THE TWX NETWORK. THESE OLDER MACHINES HANDLE THEIR OWN LIMITED SPEED CONVERSION, AND COMMUNICATIONS IS POSSIBLE BETWEEN THESE OLD MACHINES, AND THE NEWER MODELS.

ANYONE WITH A MODEL 32 (OR EQUIVALENT) COMPUTER TERMINAL SUCH AS A COMPUTER DEVICES 1102, TI SILENT DIAL, OR TERTIUM HT-20 CAN SEND TO A TWX TERMINAL. A TWX MACHINE ANSWERS WITH THE SAME "CARRIER" TONE AS A COMPUTER DOES. SO YOUR TERMINAL WILL LIGHT ITS CARRIER LIGHT (IF IT HAS ONE). EACH MODEL 32 HAS A PROGRAMMED "ANSWERBACK" UNLESS IN IT. IT IS PROGRAMMED WITH AN IDENTIFIER OF 28 CHARACTERS OR LESS. THE FIRST THREE CHARACTERS ARE USUALLY CARRIAGE RETURN (CR), LINE FEED (LF), AND TAB OUT (TO). THE LAST THREE ARE USUALLY CR, LF, CONTROL D. THE CONTROL D CHARACTER WILL TURN ON THE PAPER TAPE PUNCH OF THE DISTANT TERMINAL (IF IT IS PROGRAMMED TO DO SO, WHICH MOST ARE). THIS LEAVES 14 CHARACTERS LEFT FOR IDENTIFICATION. AN EXAMPLE WOULD BE: NMLWINGERR DET. THIS IS THE ANSWERBACK OF THE DETROIT WINGS OF THE NATIONAL HOCKEY LEAGUE, AND WOULD FURTHER

SEEM TO BE THEIR PR DEPARTMENT.

THE TWX NUMBER OF THE WINGS IS 810-221-5012. AS WE LOOK DOWN THE LISTINGS FOR DETROIT IN A 1974 OR EARLIER TWX DIRECTORY (LATER ONES LUMP A WHOLE STATE TOGETHER), WE FIND LISTINGS FOR OLDER INSTALLATIONS. THE EXCHANGES FOR THESE OLDER MACHINES ARE 312-538, 312-222, 312-342, AND 312-759. WHEN WE HIT 222-5022, WE FIND THE ANSWERING TONE OF A TWX MACHINE. OTHER TWX MACHINES IN DETROIT ARE 810-221-5127 (DETROIT POLICE), 810-221-9455 (UNIONVAL DET.), AND 810-221-5121 (MEO EXAM DET.).

DET IS OF COURSE THE CITY CODE OF DETROIT. OTHER CITY CODES ARE NYN FOR NEW YORK, LSA FOR LOS ANGELES, SFO FOR SAN FRANCISCO, AND CHG FOR CHICAGO. SMALL COMMUNITIES HAVE FOUR LETTER CITY CODES. I DON'T KNOW THE THX TRANSLATIONS FOR ALL THE AREA CITIES, BUT THOSE SENDING THX TRANSLATIONS TO TAP WILL GET A LIST OF THE FEW CITIES TRANSLATIONS ARE AVAILABLE FOR. I HAVE TRANSLATED CITIES WITH TRUCK STOPS, AND SENT MYSELF MESSAGES WHEN TRAVELING.

ANOTHER POSSIBILITY IS TO HAVE FRIENDS WITH SOMEONE WHO WORKS IN A SMALL OFFICE (OR A BRANCH OFFICE OF A LARGER COMPANY), AND USE THEIR TWX MACHINE. THE TRICK THEN IS TO CALL A TWX NUMBER THAT DOES NOT EXIST, AND LET THE PHONE COMPANY COMPUTER TELL YOU WHAT NUMBER IT THINKS YOU DIALED. THIS WORKS FOR VERY FEW CITIES.

EXAMPLE: THE WESTERN UNION TELEGRAPH CO. IN UPPER MERIDEN PIVER NJ IS 710-988-2251. IF WE DIAL 910-988-2253 ON A TWX MACHINE, WE'LL HEAR A RECORDING SAY "THE NUMBER YOU HAVE DIALED, 979-2253," WHICH TELLS US THAT BY DIALING 201-279-2251 WE CAN REACH WH SALES UPRR. IF WE THEN LOOK CLOSELY AT THE NUMBER GIVEN FOR THE FYI NEWS SERVICE OFFERED TO TWX SUBSCRIBERS, THE NUMBER GIVEN IS 310-988-5956. BY DIALING 201-279-5956 AND THROWING THE HANG-UP SET INTO THE ACOUSTIC COUPLER, OUR TERMINAL TYPES OUT "ND FYI MNR". THE ANSWERBACK OF IBM SYSTEM 7 COMPUTER LOCATED IN MAHWAH NJ. THIS MACHINE IS UPDATED ONCE AN HOUR BY UPI OVER DEDICATED RATED-UP LEASED LINES. AS THE COMPUTER SENDS ITS ANSWERBACK, IT TRANSMITS CONTROL E. CONTROL E SHOULD TRIP THE ANSWERBACK OF THE CALLING TWX MACHINE SINCE THE COMPUTER WILL GO NO

FURTHER UNTILL IT RECIEVES SOMETHING. TYPE IN ANY RANDOM CHARACTERS (NOISE), AND IT WILL LET YOU IN. TYPE CATALOG OR JUST "CAT" AS IT ONLY "FEES" THE FIRST FOUR CHARACTERS AFTER A SPACE. YOU CAN TYPE "NEWS WEAT SPOR (CR)" AND IT WILL GIVE YOU THE LATEST NEWS. WEATHER FOR 20 CITIES, AND SPORTS. THE CATALOG GIVES INSTRUCTIONS AND TELLS YOU WHAT INFORMATION IS AVAILABLE. THIS IS A "DIAL-A-TONE" TYPE SERVICE, PAID FOR BY THE LINE CHARGES OF THE LONG DISTANCE CALL IN REACHING IT.

IT SHOULD BE NOTED THAT YOUR TERMINAL SHOULD BE SET FOR 110 BAUD, AND EVEN OR NO PARITY. HAVE FUN, HAPPY PHREAKING, AND REMEMBER

TWX PR PHREAKS

## ATTN: MILO FONEBILL

Please get in touch with me through the TAP office.

Tim Edison

## COMING NEXT ISSUE !!!

Our next issue, #60, marks the 7th anniversary of TAP. To mark the occasion, issue #60 will be a special double edition with color pages. I want this to be the best fuckin' issue of TAP that we've ever put out. I'm asking all of our readers, especially those who have sat on their fat asses and done nothing to help out over the past years, to send in information, newsclips, articles, and columns. I want 4 pages just bursting with technical knowledge. Lets show Bell that we DO care!

## LETTERS FROM READERS

Dear TAP,

Perhaps some of your readers would like to chat with ships on the high seas. This is now possible through a satellite called Marisat. Actually there are two currently operational: Atlantic and Pacific, and in a very short time there will be one for the Indian Ocean too. Enclosed is a list of most of the ships that you can contact and the list continues to grow. To place a call: engage your favorite 800 or 555 number, then box off to 900-243-3640 (or 284-3090 in Conn.). This gets you the Marisat operator. She will ask you billing information, so you should have a story made up as to who (what individual/or company) is calling and give her a number to bill it to (like IBM or Continental Bakery, or whatever). She doesn't have any way (presently) of knowing where the call is coming from. (It's going to cost somebody 10 bucks a minute!) And be a good guy and give her the ship's number (like SS Jonsson #1580201) and tell her you want to speak with..... on, what the hell, the chief steward or cabin boy - anyway, the radio operator will get the call and you can bullshit with him. Sparks can tell you where the ship is, courses, speed, destination, etc.

Comsat has some other numbers that might prove fruitful if somebody knows how to exploit them: (203) 284-9030 is Data Test (associated digits 1800-922-4646, which may be a computer entry code?), 284-9240, 284-9050 (data set), and 284-9060. You might also try 284-4291/4292.

Has anybody ever "boxed" calls through TWX? I don't know if it's it's a usable system. (See article on TWX phreaking by Osbert Kilgallen in this issue of TAP.) Anyway Marisat has Telex capability too: 710-381-5920 (toll free). Additionally, AT&T long lines is 914-382-8252 and the United Nations Language bank (free translations?) is 212-686-3061.

May the force be with you,  
R2-D2

Greetings,

In Missouri, Southwestern Bell has forced itself upon an unwilling public. They now charge 20¢ per directory assistance call (after the first five free ones per month).

Well, in a burst of insight it came to me--and it has been verified by experiment and questions to operators. It is:

Call the 555-1212 number, using the 1 code, omitting the area code (as in calling directory assistance for another town in your area code). When the operator asks you "What city please?" just say the name of your town (in my case St. Louis)--and she says "go ahead." They have no way of knowing what town you are dialing from! And there is no charge for the 555-1212 number.

I, also, note that Kansas City, Mo 555-1212 info uses computers to retrieve telephone numbers. When will I be able to access by computer the up to date directory listings?

Jeff

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TAP "Ma Pa Is A Cheap Mother" T-shirts - \$4.00. Specify size and color: Medium, Large, Extra Large.

Blue, tan, or red w.

Send \$1.00 check or money order (N.Y. Cash) to:

TAP, P.O. Box 100, West 11 Street, New York, N.Y. 10001

or, P.O. Box 100, New York, N.Y. 10001

DEPT. 1, P.O. Box 100, New York, N.Y. 10001

WINGS, P.O. Box 100, New York, N.Y. 10001

## Shipboard Terminal Installations:

### An Update

Approximately 100 shipboard terminals were equipped with shipboard terminals which have been commonly used for operation with the Marisat System.

| Ship           | Country of Registration | Telephone/Telex Number | Answerback   |
|----------------|-------------------------|------------------------|--------------|
| A. Hela        | Paraguay                | 1330721                | ACWD 1330721 |
| Alexis Star    | United States           | 1501577                | AGPU 1501577 |
| American Ace   | United States           | 500101                 | AFCA 150101  |
| Aquarius       | United States           | 1500702                | AFSA 1500702 |
| Arco Anchorage | United States           | 1501134                | AFSA 1501134 |
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| Arco Anchorage | United States           | 1500934                | AFSA 1500934 |
| Arco Anchorage | United States           | 1500933                | AFSA 1500    |

